pfi

Python for Fixed Income

COURSE 3 · MARKET DATA AND TIME SERIES

Case Study: Three Years of Curve and Spreads

WEEK 2 · DAY 5

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Week 2 closes: one row, three years

Days 6 to 9 built the measures of a market screen. Today puts them in one row per series, as of one date, then uses that row to describe three years of history.

The table for a date uses only rows on or before that date.

Today's credit spreads are synthetic: invented, not ICE or any index.

Week 2: time series measures

Day 1	Changes in bp and percent
Day 2	Rolling windows, volatility
Day 3	Z-scores, percentile ranks
Day 4	Slopes and butterflies
Day 5	**Case study: three years**

summary_table: seven columns, one date

Column	Definition	From
level	the as-of value: last observation on or before the date	value_as_of, day 4
chg_1d_bp	change from the previous observation, bp	change_bp, day 6
chg_1w_bp	as-of value less the as-of value 7 days back, bp	change_since_bp, day 6
chg_1m_bp	as-of value less the as-of value 1 month back, bp	change_since_bp, day 6
zscore	against mean and sample std of 252 rows, today in	rolling_zscore, day 8
pct_rank	share of the 252 rows at or below today, percent	rolling_percentile_rank, day 8

10 Yr and 2s10s on 2 October 2026 against Excel's reference row: largest gap 4.6e-13.

Only rows on or before the as-of date

An invented later row

```
# 2026-10-05: 10 Yr 9.99, 2s10s 500 (invented)
assert_frame_equal(..., check_exact=True)
```

The table for 2 October does not change by one digit. On 5 October, the 10 Yr's one-day change is 471.0 bp.

The last-row shortcut

```
pair["10 Yr"].iloc[-1]        5.28
summary_table(..., 09-30)      5.29
```

A report for 30 September that takes today from the last row shows a value from two market days later.

Two tests: the known date against Excel, and no lookahead at three cut dates. 42 passed.

Case study: three years of curve and spreads

3 October 2023 to 2 October 2026: the window of ICE BofA's series on FRED (three years only, since April 2026). Their values may not be published, so the spreads below are synthetic.

Series	Source	Unit
3 Mo, 2 Yr, 5 Yr, 10 Yr, 30 Yr	U.S. Treasury par yield curve, public domain	percent
2s10s, 5s30s, 3m10y	slopes, long less short (day 9)	bp
2s5s10s, 5s10s30s	butterflies, 2 x belly less both wings	bp
ig_spread_bp, hy_spread_bp	SYNTHETIC: invented, not ICE or any index	bp

Every observation says what the data was. None says why it moved or what comes next.

Levels: start, end, low, high

```
levels = pd.DataFrame({
    "first": three_years[YIELDS].iloc[0],
    "last": three_years[YIELDS].iloc[-1],
    "low": ...min(), "low_date": ...idxmin(),
    "high": ...max(), "high_date": ...idxmax(),
})
# idxmin, idxmax: the FIRST date reached
```

Predict 1

Over the three years, which tenor's yield changed most from the first day to the last?

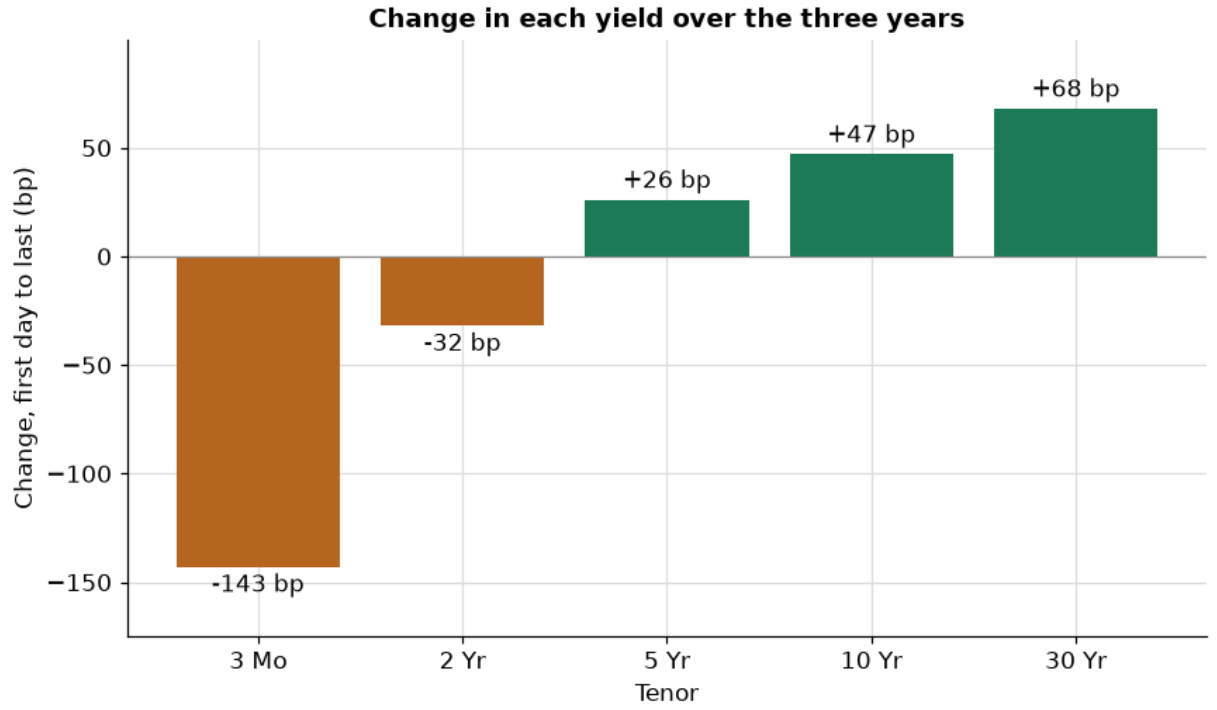
A) The 30 Yr, up

B) The 3 Mo, down

C) The 2 Yr, down

Decide, then run the cell.

Levels, drawn



Reading the picture

The bars grow from negative on the left to positive on the right as the tenor lengthens.

The 3 Mo ended lower, the 30 Yr higher. Sizes and directions only.

Daily changes in bp

```
daily = pd.DataFrame({name: md.change_bp(
    market[name], 1, units[name] == "pct")
    for name in market.columns})
daily_3y[YIELDS].max()     # largest rise
daily_3y[YIELDS].idxmin()  # date of fall
daily_3y[YIELDS].abs().mean()
# a size and a date, never a reason
```

Predict 2

On its largest single day in either direction, which moved more?

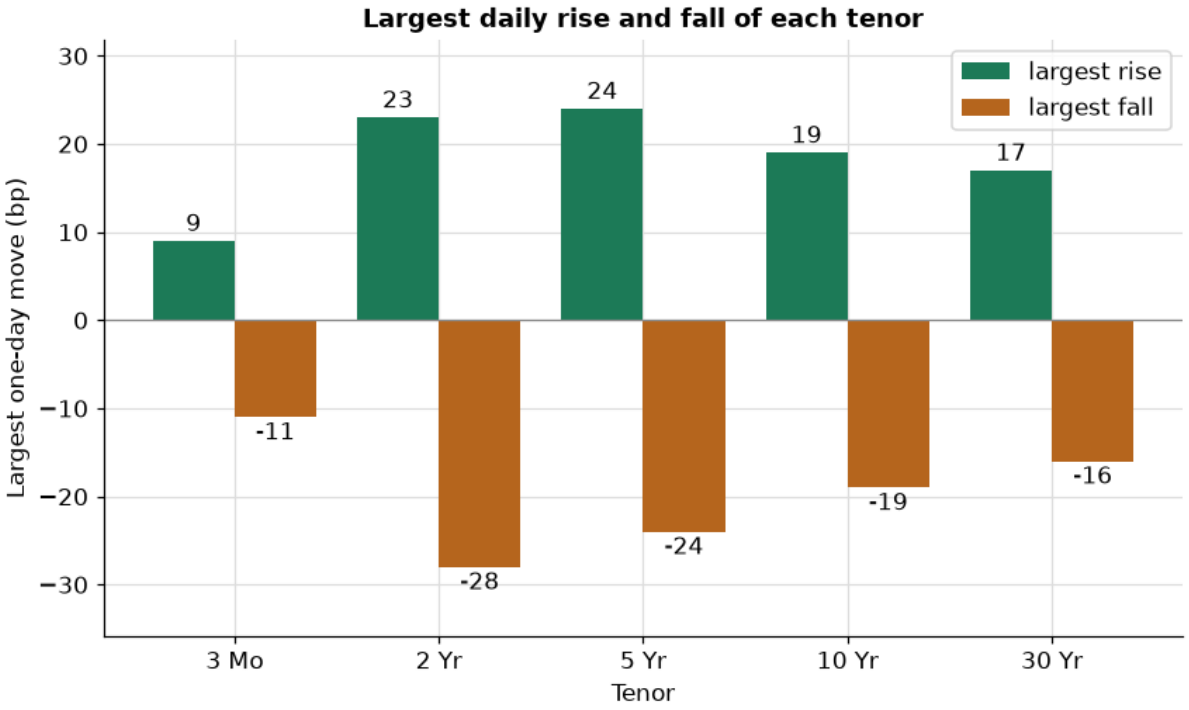
A) The 2 Yr

B) The 10 Yr

C) They moved the same

Decide, then run the cell.

Daily changes, drawn



Reading the picture

Each pair is a tenor's biggest day up and its biggest day down. The 2 Yr's pair is the tallest.

Sizes only here; the dates are in the table, and no reason is given.

Volatility regimes

```
vol_63 = pd.DataFrame({name: md.rolling_vol_bp(
    daily[name].dropna(), 63)
    for name in VOL_SERIES})
# full windows on the whole file, then
# .loc["2023-10-03":"2026-10-02"]
(vol_63 <= vol_63.iloc[-1]).mean() * 100
```

Decide, then run the cell.

Predict 3

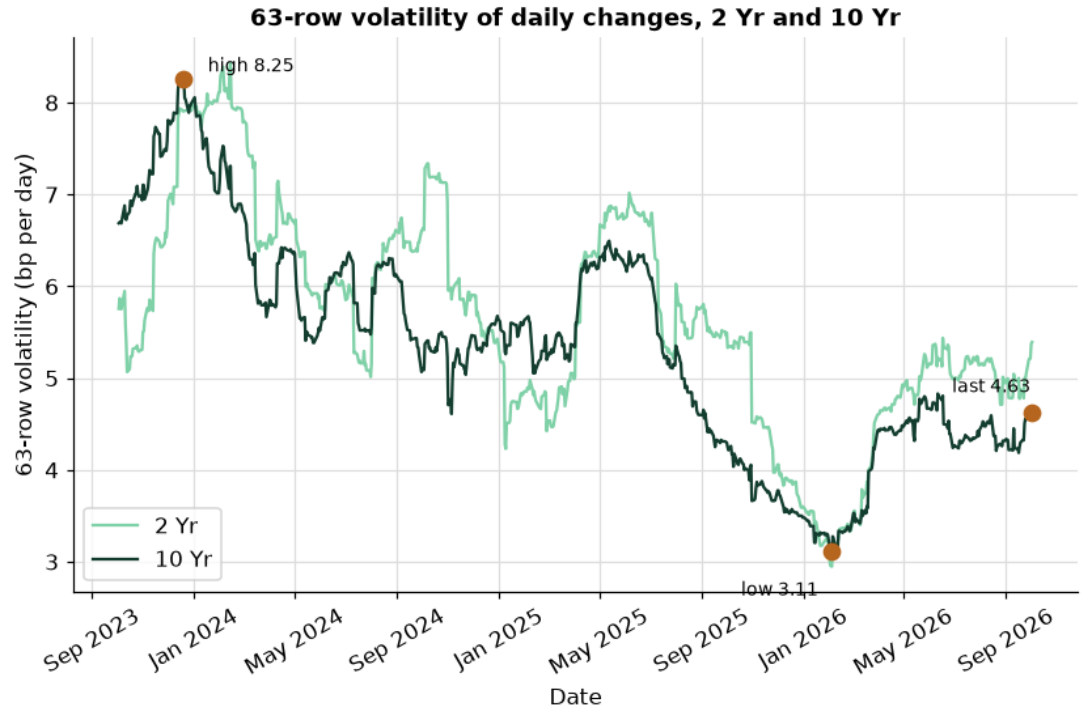
On 2 October 2026, is the 10 Yr's 63-row volatility nearer its three-year low or its high?

A) Nearer its low

B) Nearer its high

C) At its high

Volatility, drawn



Reading the picture

Dark line: the 10 Yr. Light line: the 2 Yr. The orange dots mark the 10 Yr's lowest, highest, and last values.

The last value sits near the bottom of the range.

Where each series sits on 2 October 2026

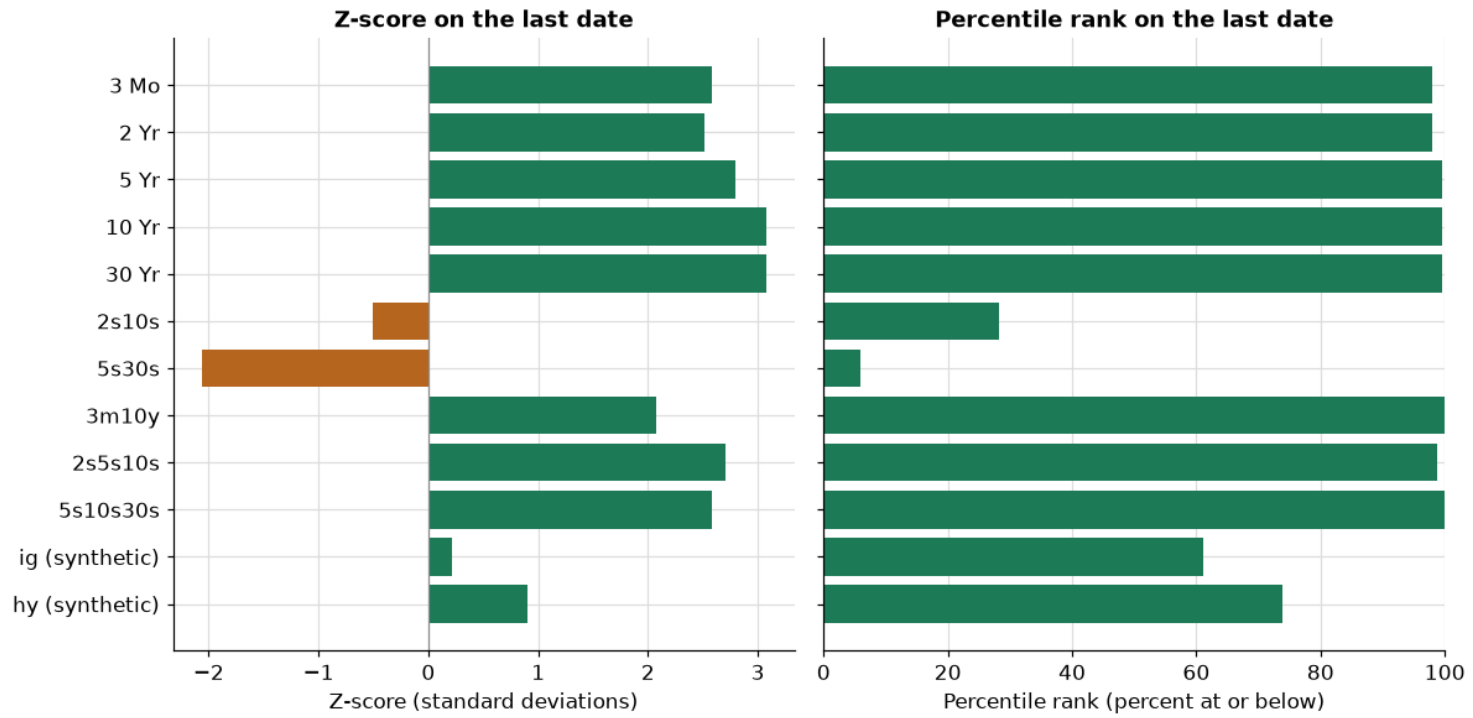
Series	Level	z-score	Pct rank
3 Mo	4.19	2.58	98.0
2 Yr	4.83	2.51	98.0
5 Yr	5.06	2.80	99.6
10 Yr	5.28	3.08	99.6
30 Yr	5.63	3.08	99.6
2s10s	45.00	-0.51	28.2

Series	Level	z-score	Pct rank
5s30s	57.00	-2.07	6.0
3m10y	109.00	2.07	100.0
2s5s10s	1.00	2.70	98.8
5s10s30s	-13.00	2.58	100.0
ig (synthetic)	97.00	0.22	61.1
hy (synthetic)	328.70	0.90	73.8

Predict 4: the 10 Yr ranks 99.6 in a window of 252 values, today's included. How many are at or below 5.28? A) 99 B) 250 C) 251

A z-score near 3 or a rank near 100 says where today sits in its past year. It never says what comes next.

Where each sits, drawn



Reading the picture

Left: each series' z-score from the mean at zero. Right: the percentile ranks on one scale from 0 to 100.

Position in each series' own past year, nothing about what comes next.

The shape of the curve

```
# 9 decimals first: binary noise is not a sign
shape = three_years[SHAPES].round(9)
"negative_days": (shape < 0).sum(),
"zero_days": (shape == 0).sum(),
"positive_days": (shape > 0).sum(),
```

Decide, then run the cell.

Predict 5

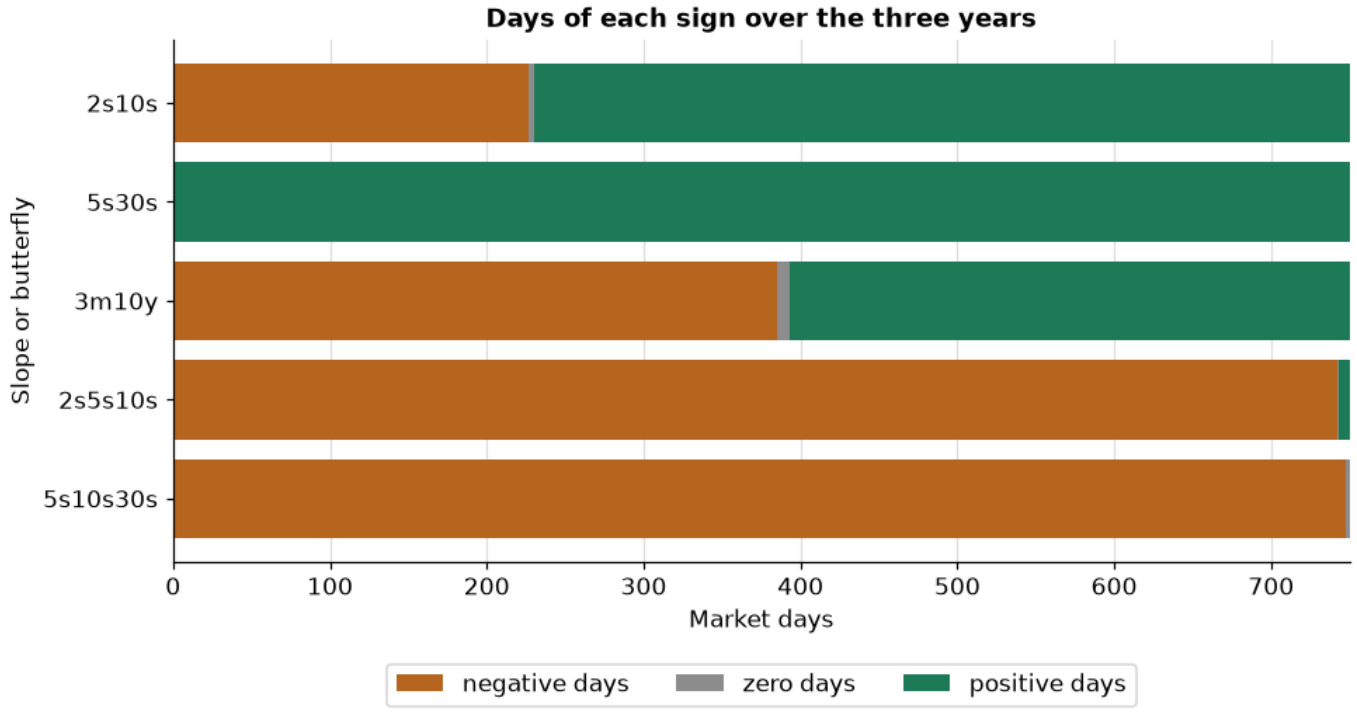
On how many of the 750 days was 2s10s negative: the 2 Yr above the 10 Yr?

A) About a tenth

B) About a third

C) About two thirds

The shape, drawn



Reading the picture

Each bar is the 750 days split by sign: orange negative, grey zero, green positive.

5s30s never negative; 5s10s30s never positive. Five separate facts.

The synthetic spreads

```
# SYNTHETIC: invented by the course, not any index
synthetic = three_years[["ig_spread_bp",
                         "hy_spread_bp"]]
gap = synthetic["hy_spread_bp"] - synthetic["ig_spread_bp"]
daily_3y["ig_spread_bp"].corr(daily_3y["hy_spread_bp"])
```

Invented series: they describe no real market.

Predict 6

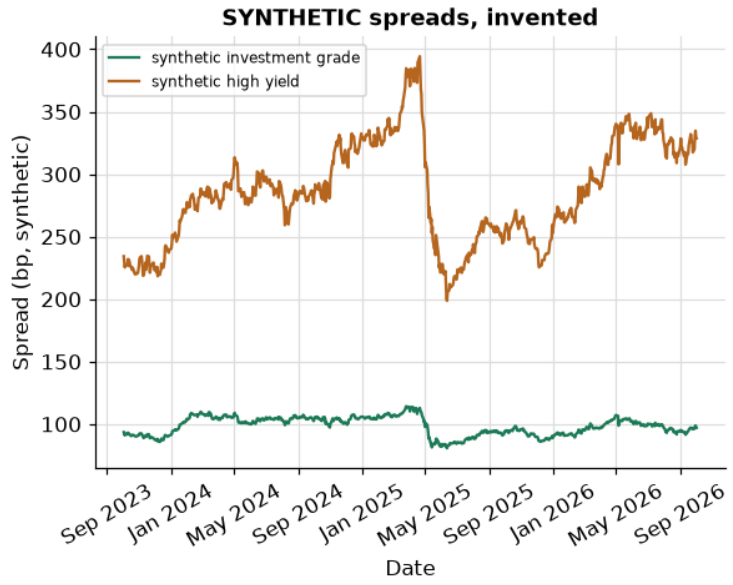
The generator builds the high yield shock from the investment grade one plus noise. Correlation of daily changes?

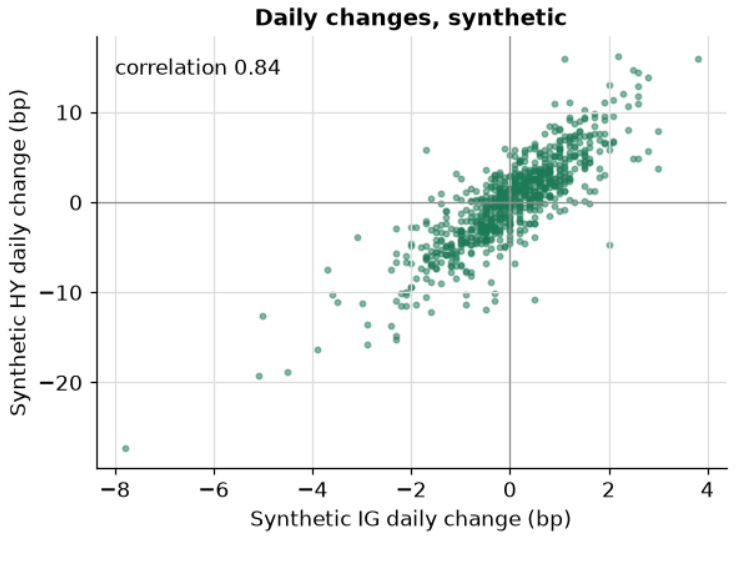
A) Closer to 0

B) Closer to 0.5

C) Closer to 1

The synthetic spreads, drawn



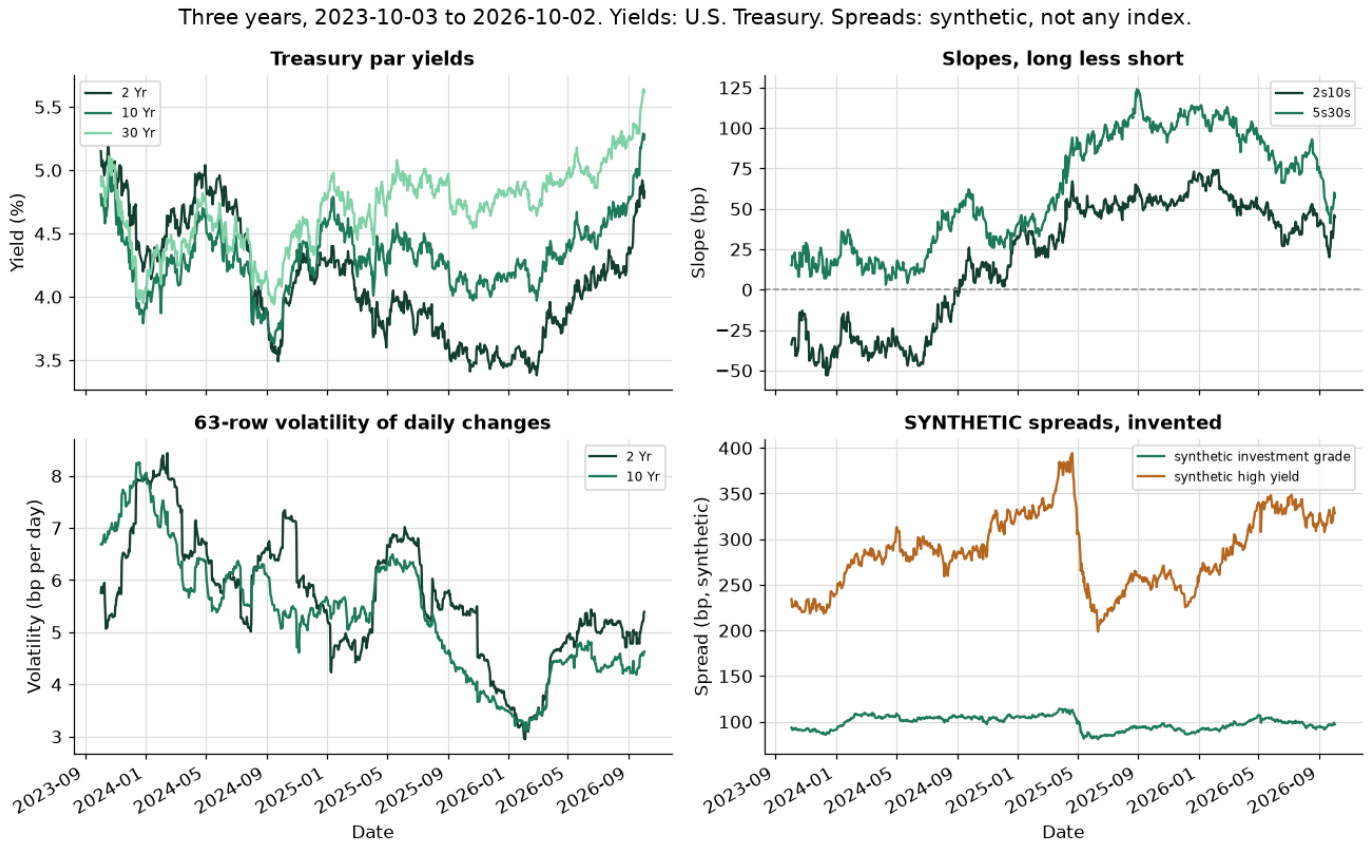


Reading the picture

Left: two invented series. Right: each dot is one day's change in both. They lie along a rising line.

Both panels are synthetic: invented, not ICE or any index.

Four panels, and observations that describe



Observations

3 Mo down, 30 Yr up over three years

Largest daily move: a 2 Yr fall

10 Yr volatility ended nearer its low

Every yield ranked 98 or higher

2s10s negative on 227 of 750 days

Synthetic spreads correlated 0.84

None says why, or what comes next.

Live: the same table from FRED

A live cell, two ways

No key (the saved notebook):

```
Live cell skipped: no FRED key was found.
```

With your key:

DGS2, DGS10: table equals the snapshot's, 1e-9

A live cell, two ways

No key (the saved notebook):

```
Live cell skipped: no FRED key was found.
```

With your key:

ICE OAS: row counts and dates only

ICE BofA's terms: not to be published or distributed. Values only under SHOW_LICENSED_VALUES, which stays False.

Recorded with no key and SHOW_LICENSED_VALUES = False: no ICE value on screen, in a notebook, or on a slide.

Excel side by side: the summary row in a sheet

```
# A dates, B 2 Yr, C 10 Yr, D 2s10s =(C2-B2)*100; as-of date in H1; row 3 = 10 Yr
H3 =XMATCH($H$1,$A$2:$A$2941,-1)
I3 =XLOOKUP($H$1,$A$2:$A$2941,C$2:C$2941,,-1)
J3 =(I3-INDEX(C$2:C$2941,H3-1))*100
K3 =(I3-XLOOKUP($H$1-7,$A$2:$A$2941,C$2:C$2941,,-1))*100
L3 =(I3-XLOOKUP(EDATE($H$1,-1),$A$2:$A$2941,C$2:C$2941,,-1))*100
M3 =STANDARDIZE(I3,AVERAGE(OFFSET(C$1,H3-251,0,252,1)),STDEV.S(OFFSET(C$1,H3-251,0,252,1)))
N3 =COUNTIF(OFFSET(C$1,H3-251,0,252,1),"<="&I3)/COUNT(OFFSET(C$1,H3-251,0,252,1))*100
```

As of 2 Oct 2026	level	1 day	1 week	1 month	z-score	pct rank
Excel, 10 Yr	5.28	4.0	11.0	49.0	3.0752	99.6032

Tested in Excel 16.0 build 20430, 2026-10-04, on three dates (one a holiday): 36 values, largest gap to summary_table 5.5e-13.

Save the day: tag week2, compare with week1

1. Rebuild week1

Cut the files at week 2's first function; commit; git tag week1

2. Commit today

summary_table and its two tests

3. Tag

git tag week2 on today's commit

4. Compare

git diff --stat week1 week2

```
marketdata.py | 216
test_marketdata.py | 229
2 files changed, 445 insertions(+)
```

marketdata.py as week 1 left it: 12 functions; test_marketdata.py: 20 tests. Week 2 only added code.

The AI check

The docstring is the prompt. The code that comes back runs, and has one data bug.

1

Docstring

Units, conventions, and what it must never do: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert what the docstring promises. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: trailing_window takes iloc[-(window + 1):], 253 rows for a 252-row window (rows: 253).

Code that runs is not the same as code that is right.

What you can do now

- Build a summary row: level, three changes in bp, z-score, rank
- Prove a table reads nothing after its as-of date
- Describe three years with numbers and dates, never reasons
- Count a sign without being fooled by binary noise
- Build the same row in Excel, and close a week with git tag

Week 3, Day 1 SQL with sqlite3: queries beside pandas, proved equal